DRUGS · ANTIFUNGALS AND ANTIVIRALS

Fluconazole

Fluconazole is a well-absorbed, water-soluble triazole with excellent activity against most Candida species but none against Aspergillus or the Mucorales. In ENT it treats oropharyngeal and oesophageal candidiasis that topical agents have not cleared or that occurs in immunocompromised or irradiated patients, candidal infection of the larynx and voice prostheses, and Candida otomycosis resistant to topical treatment.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

FLUCONAZOLE

	SUMMARY
Class	Triazole antifungal
How it works	Inhibits fungal lanosterol 14-alpha-demethylase (CYP51), depleting ergosterol; fungistatic against Candida
Spectrum	Candida albicans and most other Candida (C. krusei is resistant, C. glabrata often less sensitive), Cryptococcus; no activity against Aspergillus or Mucorales
ENT uses	Oropharyngeal candidiasis in the immunocompromised, after radiotherapy or failing topical treatment; oesophageal candidiasis; laryngeal candidiasis; candidal colonisation of voice prostheses; refractory candidal otomycosis
Adult dose	Oropharyngeal candidiasis 50 mg once daily for 7 to 14 days (100 to 200 mg in severe or immunocompromised cases); oesophageal 200 to 400 mg on day 1, then 100 to 200 mg daily for 14 to 21 days
Child dose	3 to 6 mg/kg on day 1, then 3 mg/kg daily
Pregnancy	Avoid (high doses linked to fetal malformations)

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Handling

About 90% oral bioavailability, unaffected by food or acid; long half-life (about 30 hours) allows once-daily dosing; renally excreted (halve the dose when the creatinine clearance is under 50 ml/min).

3 CHAPTER THREE Using it in ENT

Indications.

3.1 Oral and laryngeal candidiasis

Topical nystatin or miconazole first for uncomplicated thrush. Use fluconazole for extensive disease, failure of topical treatment, during or after radiotherapy, in HIV and other immunosuppression, in steroid inhaler users with laryngeal candidiasis (with advice to rinse after inhaler use), and for oesophageal involvement (odynophagia).

3.2 Voice prostheses

Candida biofilm shortens the life of tracheo-oesophageal voice prostheses; antifungal treatment (oral or topical) can extend device life in patients with frequent leaks.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

Nausea, abdominal pain, rash, headache; hepatotoxicity (rare); QT prolongation; Stevens-Johnson syndrome rarely.

4.2 Interactions

A moderate CYP3A4 and strong CYP2C9 and 2C19 inhibitor: raises **warfarin** (INR), phenytoin, sulfonylureas, ciclosporin, tacrolimus and some statins; additive QT effects with other QT-prolonging drugs.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Spectrum: Candida yes, **Aspergillus and Mucor no**.
- Use in head and neck radiotherapy and voice prosthesis patients.
- Warfarin and QT interactions.

5.2 Common pitfalls

COMMON PITFALLS

- Using fluconazole for suspected invasive fungal sinusitis.

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